

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000841	Version:	4.0, CURRENT
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DG-57 / DG-517 Service Manual

LUBRICATION POINTS AND PROCEDURES

Procedure description
This procedure describes how to lubricate several points.

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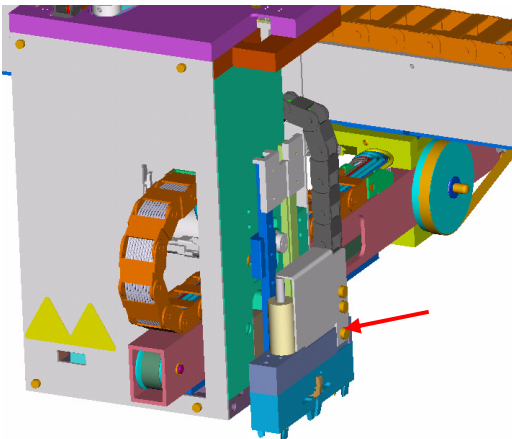
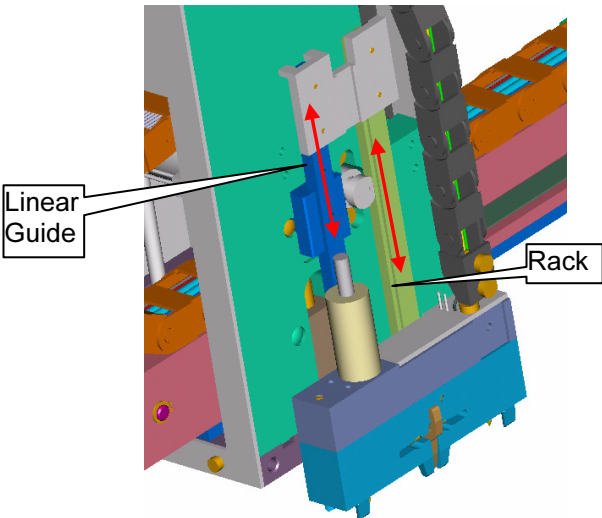
EQUIPMENT

1. 7 ml dispenser of oil *Super Lube* Synthetic lubricant with syncolon (PTFE). (Cod. 232696).
2. Tube of 85 g. of grease *Super Lube* Synthetic grease with syncolon (PTFE). (Cod. 232697).

SET UP

1. Turn off the instrument with the door opened.

PROCEDURE

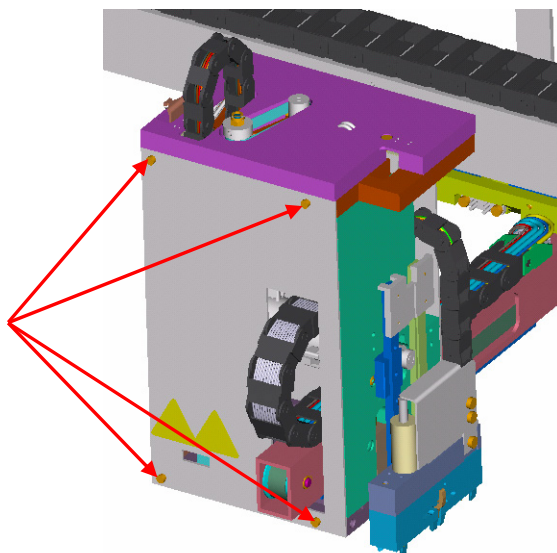
Process	Expected Results
1. Gripper guide system. Remove the PCB protection of the gripper unscrewing the bolt. Is not necessary to remove completely.  Picture 1 Grease the both sides of linear guide and the toothed part of the rack. Move manually up and down several times.  Picture 2 Replace the PCB protection screwing the bolt (Picture 1).	---

Process

Expected Results

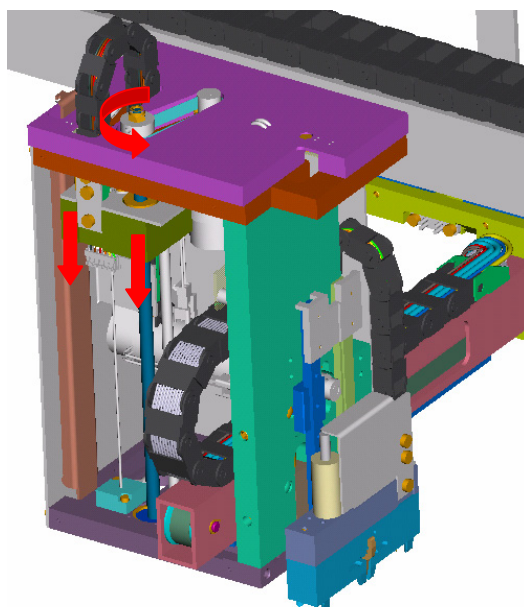
2. Leadscrew and wagon guide.

Remove the front cover of the arm unscrewing the 4 bolts.



Picture 3

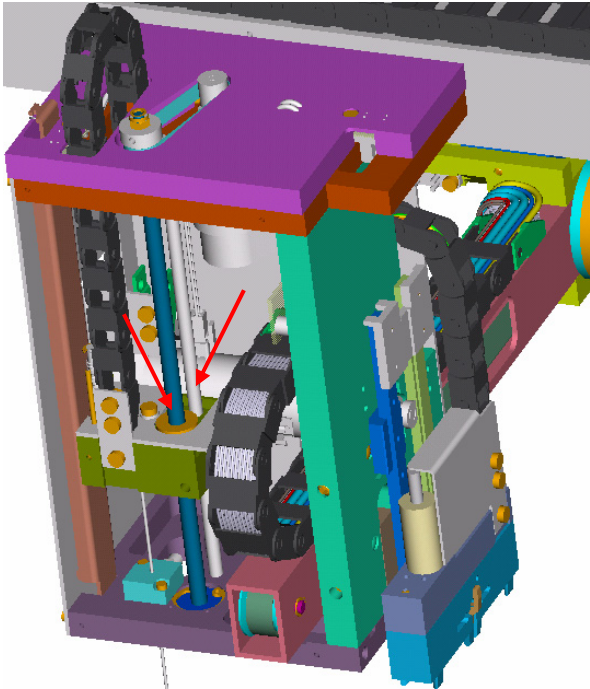
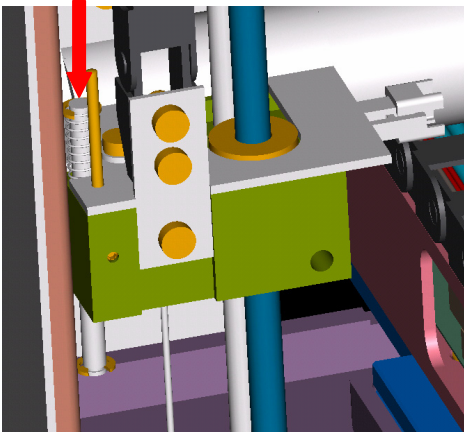
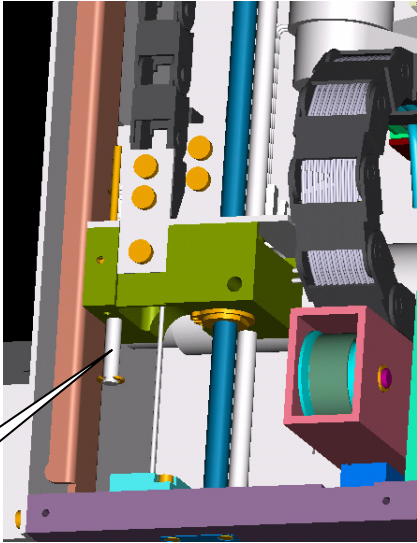
Move the arm to a position than is possible to lower the probe.

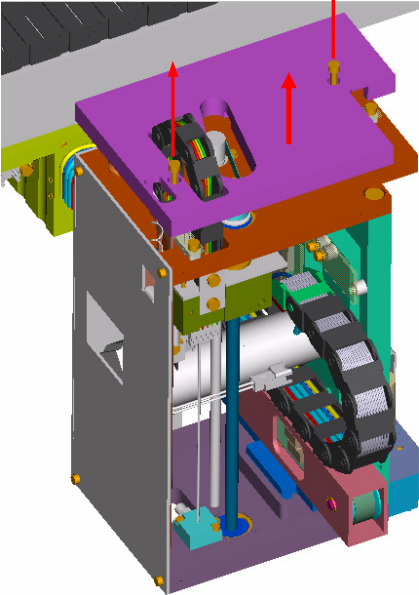
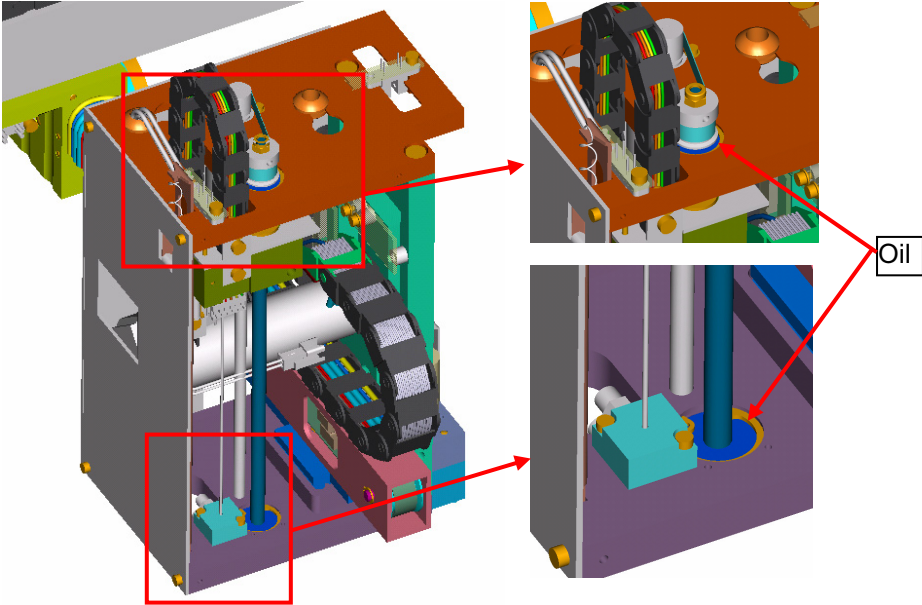


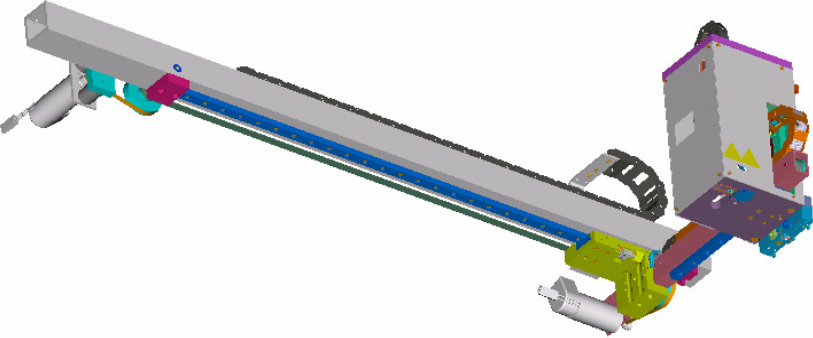
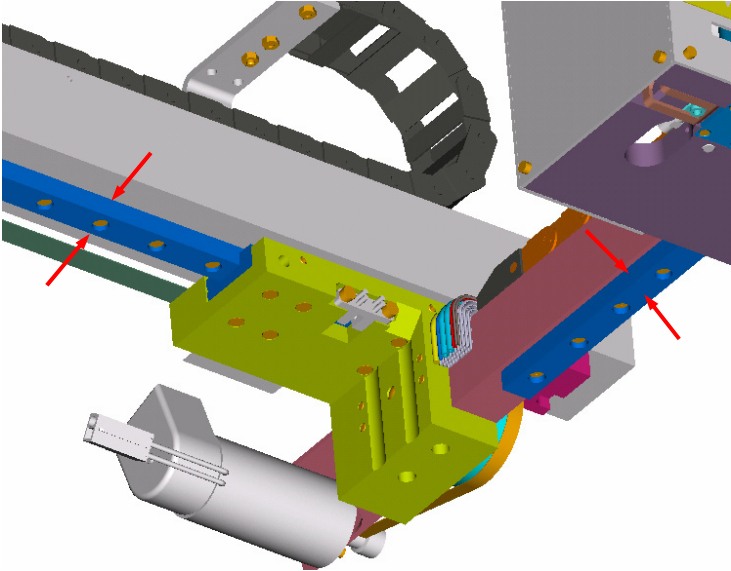
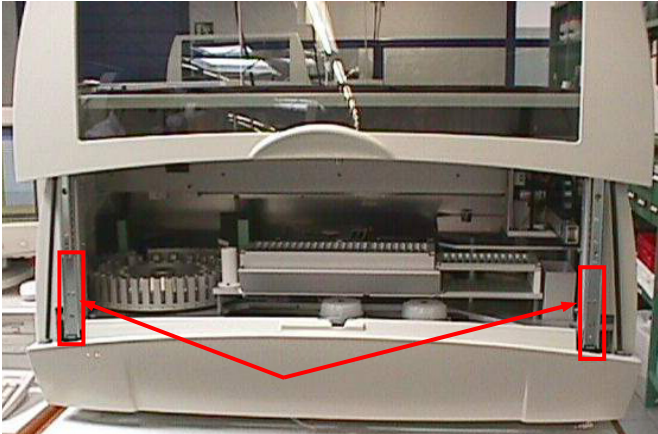
Picture 4

Lower manually the probe as much as possible.

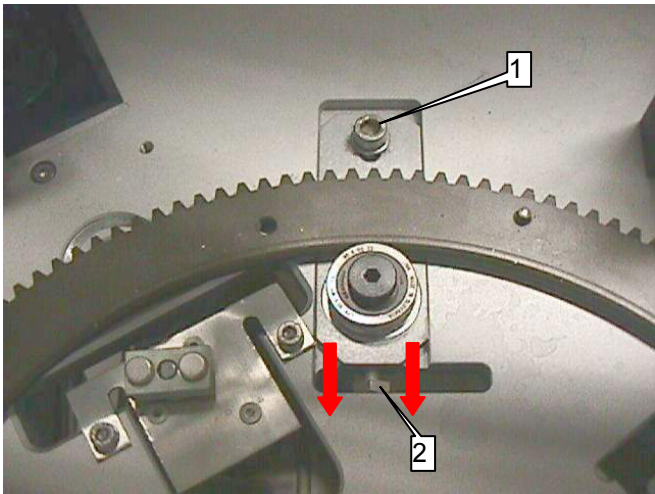
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Process	Expected Results
Grease both leadscrew and wagon guide just over the wagon (low quantity).  Picture 5 Move up and down the wagon several times. Grease the spring guide of the cards treader. Place the wagon in the same position than in the Picture 5. Push down the spring and grease the lower part of the guide while the guide is been pushed.   Picture 6 Picture 7	

Process	Expected Results
Lubrication of leadscrew ball bearings. Remove the top cover of the arm unscrewing the two bolts.  Picture 8 With the oil dispenser put a drop of oil in the inner part of the ball bearing:  Picture 9 Move the probe up and down several times. Place the probe in the upper position and replace the top cover (Picture 8) and the front cover (Picture 3). 3. Grease the X and Y guides. Place the arm to the maximum right position (X direction) and move in the Y direction to the outer position. Grease with small amount of both sides of X and Y linear guide.	---

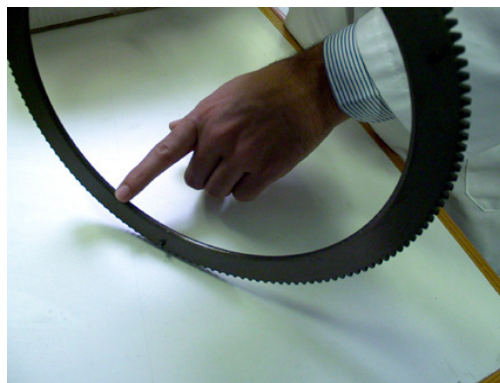
Process	Expected Results
 Picture 10  Picture 11 Move from end to end the in X and Y directions several times. 4. Grease the door guide and the door lock: Grease only the inner part of the guides marked in the Picture 13. Grease the guides (the fixed part screwed to the instrument chassis).  Picture 12	---

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Process	Expected Results
 Picture 13 Grease the end (round part) of the door latch.  Picture 14 Open and close the door several times. 5. Grease the samples carrousel crown: Remove the samples and reagents carrousels from the instrument. Screws the tension adjust screw of the tension bearing (1 in picture 15). Unscrew the fixing screw of the tension bearing. Is not necessary to extract the screw. (2 in picture 15). Move the tension bearing to extract the crown.  Picture 15	---

Process**Expected Results**

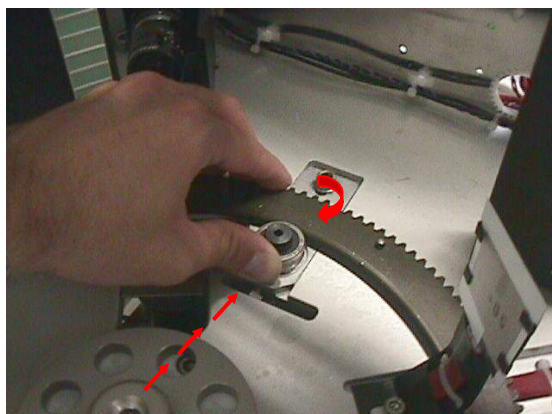
Grease the inner side of the crown with a small layer of grease.



Picture 16

Insert the crown in its position.

Tight manually the tension bearing (by pushing it with the hand) and tight (not firmly) the fixing screw. The strength applied is not critical as long as crown keeps held by the three bearings. Normal force applied by hand, as shown in the picture 17, is enough.

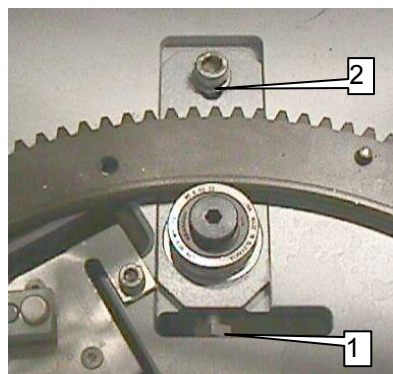


Picture 17

Move manually the crown 360° (a whole turn).

If higher strength is required to move it in one point, then adjust again manually the tension bearing at this point, that's to say, repeat the previous step at the point of maximum friction.

Unscrew the tension screw (1 in picture 18) until it touches the basis. Tight it 1/8 of a turn more.



Picture 18

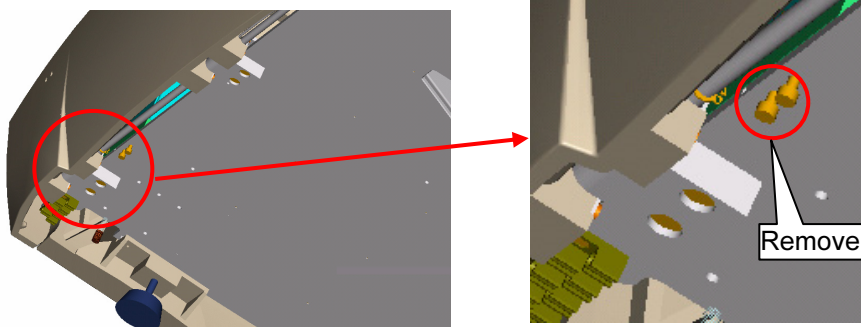
Process**Expected Results**

Tight firmly the fixing screw. (2 in picture 18).

Check again the movement of the crown by rotating the crown manually. If it does not work properly repeat all the procedure.

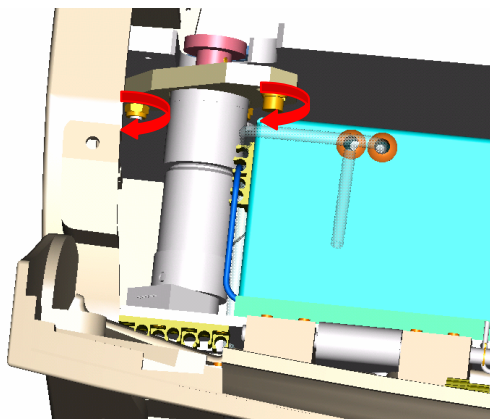
Once the tension of the crown is adjusted, the distance between the gearmotor and the crown has to be adjusted.

With the front bottles door closed unscrew and remove the fixing screws of the pumps box, located in the bottom part of the instrument.



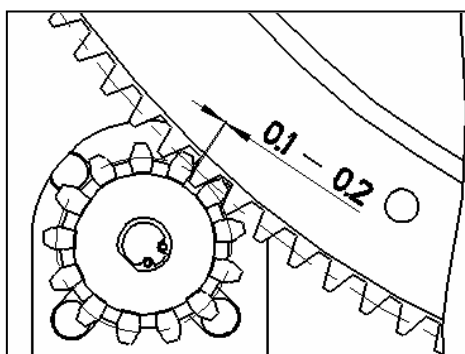
Picture 19

Move slightly the pumps box to allow the access of the screws showed in picture 20.



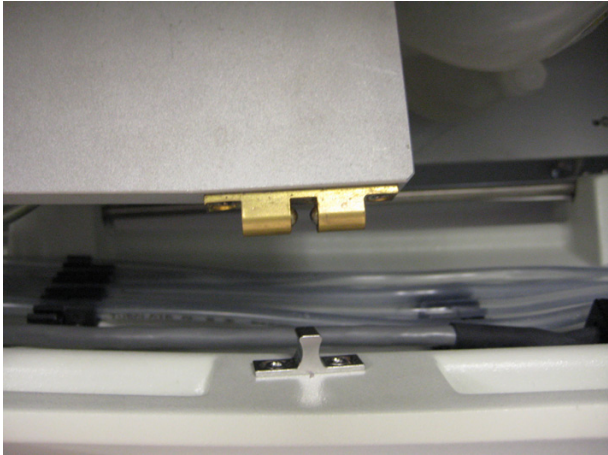
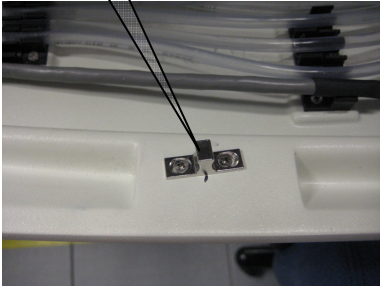
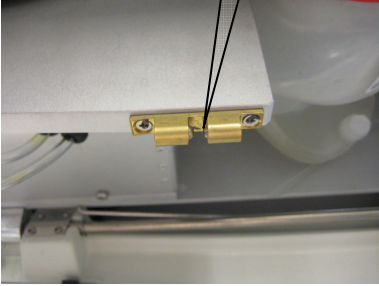
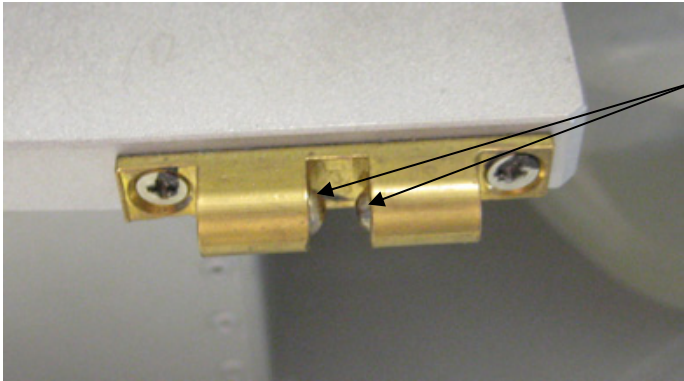
Picture 20

Loose the two screws to allow the movement of the position of the motor. Move the motor. The distance between the crown and the gear has to be between 0.1 and 0.2 mm.



Picture 21

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Process	Expected Results
Tight the two screws those hold the motor to the instrument basis. Replace the pumps box following the procedure REGD-0000829 from the step 17. Close the door. Follow the Procedure 3 from REGD-0000797 procedure to check the carrousel. 6. Grease the Bottle Door Latch with <i>Super Lube</i> Synthetic grease with syncolon (PTFE). (Cod. 232697). Open the bottle door:  Grease both parts of the bottle door latch, see the following pictures:   The two balls of the female bottle door latch shall be greased:  Open and close the bottle door several times.	---

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DOCUMENT CHANGE HISTORY

Version	Date	Change	Rationale
2.0	2005/05/09	NA	Replaced document 57070V0M000.
3.0	2009/07/02	Added the lubrication of the Bottle Door Latch.	Due to the change of the bottle door latch material. See document REGD-0004784.
4.0	See document header	Document REGD-0000841_v4 replaces document REGD-0000841_v5.	Document version changed.
4.0	See document header	Date format changed to YYYY/MM/DD inside document change history.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	See document header	Document title.	Commercial reference substituted by project reference.
4.0	See document header	Header title.	Increase information to easy document search.